Fourth-moment comparison of zero-field SK free-energy fluctuations

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Abstract

We compare zero-field Sherrington–Kirkpatrick log partition functions with independent, possibly nonidentically distributed, centered variance-one couplings under a uniform fourth-moment bound. The mean error is bounded independently of system size, the variance error is bounded by a constant times one plus the Gaussian standard deviation, and bounded smooth tests at scale $a \geq 1$ have error $O(a^{-1})$. Third moments need not vanish. The cubic term in coordinate replacement is controlled by a second replacement over a reflected vertex star. An expansion of the square and a finite family of hybrid laws give the variance estimate without fifth moments. Assuming the explicitly identified Gaussian low-temperature theorem, these comparisons transfer its exact variance prefactor and its limiting law with each model's own mean and variance. The Gaussian theorem is an input, not a result proved here.

1 Statement, inputs and antecedents

For $n \geq 2$, put

$$E_n = \{\{i,j\} : 1 \leq i < j \leq n\}, \qquad m = |E_n| = \frac{n(n-1)}{2}, \qquad q = \frac{\beta}{\sqrt{n}}.$$

For $x \in \mathbb{R}^{E_n}$ define the counting-prior, zero-field log partition function

$$F(x) = \log \sum_{\sigma \in \{-1,1\}^n} \exp\left(q \sum_{e=\{i,j\} \in E_n} x_e \sigma_i \sigma_j \right). \tag{1}$$

Let $g = (g_e)$ have independent standard Gaussian coordinates and write $c_G = \mathbb{E}F(g)$ and $V_G = \operatorname{Var} F(g)$. Throughout, the temperature is fixed.

Theorem 1.1 (Finite-size comparison)**.** *For every $\beta > 0$ and $K \geq 1$ there is $C = C(\beta, K) < \infty$ such that the following holds for every $n \geq 2$. Let $(J_e)_{e \in E_n}$ be independent real variables with*

$$\mathbb{E}J_e = 0, \qquad \mathbb{E}J_e^2 = 1, \qquad \mathbb{E}|J_e|^4 \leq K. \tag{2}$$

Then

$$|\mathbb{E}F(J) - c_G| \leq C, \tag{3}$$

$$|\operatorname{Var} F(J) - V_G| \leq C(1 + \sqrt{V_G}). \tag{4}$$

For every $a \geq 1$, $c \in \mathbb{R}$, and real $\phi \in C^6(\mathbb{R})$ whose derivatives of orders zero through six are bounded,

$$\left|\mathbb{E}\phi\Big(\frac{F(J)-c}{a}\Big) - \mathbb{E}\phi\Big(\frac{F(g)-c}{a}\Big)\right| \leq \frac{C}{a}\sum_{r=1}^{6}\left\|\phi^{(r)}\right\|_{\infty}. \tag{5}$$

The same constant is independent of n, the individual coordinate laws, a, c and ϕ. No symmetry, identical-distribution, zero-third-moment, fifth-moment or tail-shape assumption is required.

The finite-size theorem is proved below without an upstream probabilistic theorem. Its low-temperature consequence uses a separate Gaussian input. We state that input explicitly rather than deriving it from the fluctuation exponent alone.

Proposition 1.2 (Assumed Gaussian input)**.** *Accept the limiting-law theorem and moment theorem of the pinned manuscript [5], at the revision specified in the bibliography. For every fixed $\beta > 1$, they supply $a_2(\beta) > 0$ and a probability law ν_β such that*

$$\frac{V_G}{n^{1/3}} \longrightarrow a_2(\beta), \qquad \frac{F(g)-c_G}{\sqrt{V_G}} \Longrightarrow \nu_\beta \tag{6}$$

as $n \to \infty$ through all integers. The law has mean zero and variance one.

The model in [5, `eq:model`] is exactly (1). That source's moment theorem concerns

$$F_n^c = F(g) - n\log 2 + \zeta_n, \qquad Y_n = n^{-1/6}(F_n^c - \mathbb{E}F_n^c),$$

where ζ_n is independent centered Gaussian with variance $\beta^2/2$. Its second-moment limit $a_2 > 0$ implies

$$n^{-1/3}V_G = \operatorname{Var} Y_n - \frac{\beta^2}{2n^{1/3}} \longrightarrow a_2.$$

Its limiting-law theorem uses the uncompleted model's exact mean and variance; the completion-removal argument is also given in [6, `red:prop-moments`]. The companion [7] supplies only the exponent $n^{1/3+o(1)}$, which would not by itself identify the prefactor or limiting law. These upstream assertions are stipulated inputs here; their proofs are not audited or reproduced.

Corollary 1.3 (Conditional low-temperature transfer)**.** *Assume Proposition 1.2. For every fixed $\beta > 1$ and every triangular array of independent edge families satisfying (2) with one finite uniform K,*

$$\frac{\operatorname{Var} F(J)}{n^{1/3}} \longrightarrow a_2(\beta), \qquad \frac{F(J)-\mathbb{E}F(J)}{\sqrt{\operatorname{Var} F(J)}} \Longrightarrow \nu_\beta. \tag{7}$$

The denominator is positive for all sufficiently large n, and convergence is through all integers. No coupling between different sizes is required.

Contribution and antecedents. Carmona–Hu [1] use approximate integration by parts and interpolation to compare free energies; their Section 4.2 obtains an absolute $O(1)$ comparison with a vanishing third moment and a finite fourth moment. Coordinate replacement and bounded-spin log-partition derivative estimates are classical; see Chatterjee [2]. Cheng–Liu–Shao–Xu [3, Proposition 3.2] give a particularly close scaled smooth-test comparison under matching of the first three moments.

There is an earlier conceptual precedent for controlling a mismatched cubic coefficient by an extra cancellation in its expectation. In edge Green-function comparison, Erdős–Yau–Yin [4, end of

Section 6.1 and Lemmas 6.5–6.6] expand after deleting a matrix row and column; local-law estimates isolate a centered linear term independent of the minor. That principle is credited here. Our implementation for SK uses deterministic gauge oddness and comparison with a reflected product law, together with uniform Gibbs cumulant bounds. It does not import a resolvent expansion or local-law estimate.

The contribution considered here is the integrated comparison (3)–(5) with unrestricted third moments. Gauge symmetry and ordinary matched-moment replacement are antecedents. The square-remainder identity and finite-maximum bootstrap are a technical part of the argument, not a separate contribution. The law and prefactor in Corollary 1.3 are inherited from the Gaussian input. This is a bounded comparison with the primary operations cited above, not a worldwide priority assertion. No external-field, varying-temperature, critical-window or ground-state extension is claimed.

2 Derivatives and the vertex gauge

Write $s_e(\sigma) = \sigma_i\sigma_j$ for $e = \{i, j\}$. For an ordered list $\boldsymbol{e} = (e_1, \dots, e_k)$, repetitions allowed, let $D_{\boldsymbol{e}}$ denote the corresponding partial derivative.

Lemma 2.1 (Uniform mixed derivatives)**.** *For each $k \geq 1$ there is $B_k < \infty$, depending only on k, with*

$$|D_{\boldsymbol{e}}F(x)| \leq B_k q^k \qquad (x \in \mathbb{R}^{E_n}). \tag{8}$$

In particular $|\partial_e F| \leq q$. For $U(x) = \phi((F(x) - c)/a)$ as in Theorem 1.1, put $A_\phi = \sum_{r=1}^{6} \left\|\phi^{(r)}\right\|_\infty$. Then

$$|D_{\boldsymbol{e}}U(x)| \leq C_k q^k a^{-1} A_\phi, \qquad 1 \leq k \leq 6. \tag{9}$$

Proof. Under the finite Gibbs law at x, $D_{\boldsymbol{e}}F$ is q^k times the joint cumulant of $s_{e_1}, \dots, s_{e_k}$. Its exact partition formula is

$$\kappa(s_1, \dots, s_k) = \sum_{\pi \in \mathcal{P}([k])} (|\pi| - 1)!(-1)^{|\pi|-1} \prod_{A \in \pi} \mathbb{E}_x \prod_{r \in A} s_r.$$

Every spin product has absolute value one, so take $B_k = \sum_{\pi \in \mathcal{P}([k])} (|\pi| - 1)!$. No distinctness of the indices is used. The first derivative is the Gibbs mean of qs_e, giving its sharper bound q.

For the composition, the multivariate chain rule gives

$$D_{\boldsymbol{e}}U = \sum_{\pi \in \mathcal{P}([k])} a^{-|\pi|} \phi^{(|\pi|)}((F - c)/a) \prod_{A \in \pi} D_{(e_r : r \in A)}F.$$

The product has total derivative order k. Since $a \geq 1$ and $|\pi| \geq 1$, $a^{-|\pi|} \leq a^{-1}$, proving (9) with no dependence on c. □

The finite sum also gives the elementary growth bound

$$|F(x)| \leq n \log 2 + q \sum_{e \in E_n} |x_e|. \tag{10}$$

Thus every product law used below has finite first and second free-energy moments.

Fix a vertex i and let T_i negate all incident edge coordinates. Negating σ_i in the spin sum gives

$$F(T_i x) = F(x). \tag{11}$$

This is a deterministic identity, not invariance of an asymmetric disorder law. It holds also for $U = \Psi(F)$. For $e = \{i, j\}$, let S be the other incident edges, so $|S| = n-2$. Differentiate (11) three times in x_e before setting $x_e = 0$. For

$$h(x_{E_n \setminus \{e\}}) = \partial_e^3 U(x)\Big|_{x_e=0}$$

we obtain

$$h(-x_S, x_{E_n \setminus (S \cup \{e\})}) = -h(x_S, x_{E_n \setminus (S \cup \{e\})}). \tag{12}$$

For $n = 2$ the star is empty, and this identity states $h = 0$.

3 Two replacements: means and bounded tests

Put $K_0 = \max(K, 3)$. Consider product laws that select independently at each edge one of

$$\text{law}(J_e), \quad \text{law}(-J_e), \quad N(0,1). \tag{13}$$

Their coordinate fourth moments are at most K_0, third absolute moments at most $K_0^{3/4}$, and first absolute moments at most one. All means are zero and variances one.

In this section let $U = F$ with $L = 1$, or let $U = \phi((F-c)/a)$ with $L = a^{-1}A_\phi$. The derivative bound in either case is $C_k q^k L$ through order six.

Lemma 3.1 (Reflected-star comparison)**.** *Under every product law from (13), with e deleted,*

$$\left| \mathbb{E}\, \partial_e^3 U \Big|_{x_e=0} \right| \le Cnq^6 L, \tag{14}$$

where C depends only on K and absolute derivative constants.

Proof. Keep the exterior coordinates fixed. Compare the star product law P with the product law P^- in which each star coordinate is reflected, replacing the coordinates of S one at a time. For a replaced coordinate f, Taylor-expand h at $x_f = 0$ through degree two. Its remainder is at most

$$Cq^6 L |x_f|^3,$$

because $\partial_f^3 h$ is a mixed sixth derivative. The first two moments of a law and its reflection agree, so the polynomial terms cancel. Each replacement costs at most $Cq^6 L K_0^{3/4}$ after expectation. Averaging the exterior does not change this bound. The endpoints have opposite expectations by (12); dividing their difference by two proves (14). Every intermediate law is a product law. In particular, no shared random sign is used to create dependent coordinates. □

Proof of (3) and (5). Enumerate the edges and replace any product law in (13) by the Gaussian law one coordinate at a time. At edge e, condition on the remaining coordinates and write $z = x_e$. Taylor expansion gives

$$U(z) = U(0) + z\partial_e U(0) + \frac{z^2}{2}\partial_e^2 U(0) + \frac{z^3}{6}\partial_e^3 U(0) + R(z), \qquad |R(z)| \le Cq^4 L |z|^4.$$

The constant, linear and quadratic terms cancel. The cubic difference is the third-moment difference times the expectation of the deleted-edge cubic derivative, divided by six: the replaced variable is

independent of that coefficient. Lemma 3.1 and the moment bounds show that one replacement costs at most $CL(q^4 + nq^6)$. Exactly,

$$mq^4 = \frac{\beta^4(n-1)}{2n} \le \frac{\beta^4}{2}, \qquad mnq^6 = \frac{\beta^6(n-1)}{2n} \le \frac{\beta^6}{2}. \tag{15}$$

Summing proves a bound $C(\beta, K)L$. For $U = F$, integrability follows from (10). The mean comparison is proved uniformly for every hybrid law, which will be used below. For the test observable it is exactly (5). The sixth derivative in the inner comparison is integrated against a third power of its own coordinate; it is not a sixth-moment assumption. □

4 Variance with only fourth moments

4.1 A finite maximum and deleted coordinates

Let $\mathcal{H}$ be the finite class of all product laws in (13), with at most 3^m members. It is closed under every outer replacement and star reflection used above. Define

$$M = \max_{P \in \mathcal{H}} \mathbb{E}_P (F - c_G)^2. \tag{16}$$

This maximum is finite before any variance estimate: (10) and finite second moments suffice. No bound depending on 3^m will be used.

If $r \le 2$ coordinates of a hybrid are set to zero, restore them independently with compatible laws from $\mathcal{H}$. The restored array has a law in $\mathcal{H}$, so coordinate Lipschitz continuity and Cauchy–Schwarz give

$$\mathbb{E}|F_{\text{deleted}} - c_G| \le \sqrt{M} + rq. \tag{17}$$

Indeed, pointwise the left-hand absolute value is at most the restored absolute value plus q times the sum of the restored coordinate magnitudes. This estimate is unconditional, after the exterior coordinates are averaged. We do not claim a conditional bound by $\sqrt{M}$ at each fixed exterior realization.

4.2 The square remainder

Fix an edge e and its exterior coordinates. Write

$$H(z) = F(x_e = z) - c_G = H_0 + u(z), \qquad u(0) = 0, \qquad b_j = u^{(j)}(0).$$

All derivatives of positive order satisfy $|u^{(k)}| \le B_k q^k$, and $|b_1| \le q$. Taylor's theorem gives

$$\left| u(z) - b_1 z - \frac{b_2 z^2}{2} - \frac{b_3 z^3}{6} \right| \le C q^4 |z|^4. \tag{18}$$

The following exact identity avoids a fifth moment:

$$u(z)^2 - b_1^2 z^2 - b_1 b_2 z^3 = 2 b_1 z \left(u(z) - b_1 z - \frac{b_2 z^2}{2} \right) + (u(z) - b_1 z)^2. \tag{19}$$

The bracket in the first term is $O(q^3|z|^3)$, and the last bracket is $O(q^2|z|^2)$. Thus the absolute value of (19) is at most $Cq^4|z|^4$. Expanding $H^2 = H_0^2 + 2H_0 u + u^2$ gives its degree-three Taylor polynomial with remainder

$$|R_{\text{sq}}(z)| \le C q^4 |z|^4 (|H_0| + 1). \tag{20}$$

Under a product law, H_0 is independent of z. By (17) with one deletion,

$$\mathbb{E}|R_{\mathrm{sq}}(z)| \le Cq^4 K_0(\sqrt{M}+q+1). \tag{21}$$

This argument does not bound a fourth derivative of H^2 at a random interpolation point by $|H_0|+q|z|$ and then multiply it by $|z|^4$; that route would unnecessarily introduce a fifth moment.

4.3 The cubic square coefficient

With e set to zero, put

$$B = \partial_e^3[(F-c_G)^2] = 2(F-c_G)\partial_e^3 F + 6(\partial_e F)(\partial_e^2 F). \tag{22}$$

Equation (12) applies to this observable too. To control its star replacement, set $H = F - c_G$ before restriction. For any other incident edge f the full product formula is

$$\partial_e^3\partial_f^3(H^2) = \sum_{a,b=0}^{3} \binom{3}{a}\binom{3}{b}(\partial_e^a\partial_f^b H)(\partial_e^{3-a}\partial_f^{3-b}H). \tag{23}$$

Only the pairs $(a,b) = (0,0), (3,3)$ retain an undifferentiated H; they contribute $2H\partial_e^3\partial_f^3 F$. Every other term is a product of positive-order derivatives with total order six. Lemma 2.1 therefore gives

$$|\partial_f^3 B| \le Cq^6(|F-c_G|+1), \tag{24}$$

including the repeated occurrences of e and f.

For the inner Taylor variable $z = x_f$ with e still zero, Lipschitz continuity gives $|F(z)-c_G| \le |F(0)-c_G| + q|z|$. The order-two integral Taylor remainder for B is consequently at most

$$Cq^6[(|F(0)-c_G|+1)|z|^3 + q|z|^4]. \tag{25}$$

Here $F(0)$ has exactly e and f deleted and is independent of z. Use (17) with two deletions, then the third and fourth moment bounds. A comparison of the law of z with its reflection cancels the polynomial through degree two and costs at most

$$Cq^6[(\sqrt{M}+2q+1)K_0^{3/4} + qK_0] \le C_{\beta,K}q^6(\sqrt{M}+1),$$

since $q \le \beta/\sqrt{2}$. At each step the remaining coordinates have independent hybrid laws. The coefficients of the canceled polynomial have at most linear growth in the twice-deleted $|F(0)-c_G|$, so their expectations are finite. Formula (25) is integrable as well; conditional cancellation may therefore be averaged legitimately.

The two endpoint expectations of B are opposite by gauge oddness. Telescoping the $n-2$ star coordinates gives

$$|\mathbb{E}B| \le C_{\beta,K}nq^6(\sqrt{M}+1). \tag{26}$$

Only a third and a fourth moment have been used in this inner replacement.

4.4 Closing the bootstrap

Proof of (4). Telescope $(F - c_G)^2$ from an arbitrary $P \in \mathcal{H}$ to the Gaussian law. Its degree-three expansion has cubic coefficient $B/6$. The degree-zero, one and two terms cancel in expectation. Independence of the replaced coordinate and its deleted base permits multiplication of its third-moment difference by $\mathbb{E}B$. Bounds (21) and (26) show that one edge costs at most $C_{\beta,K}(q^4 + nq^6)(\sqrt{M} + 1)$. The polynomial terms are integrable: the constant term is integrable because H_0 has finite second moment, and other coefficients have at most linear growth in a deleted free energy, independently of the replacing variable. Summing with (15) yields, for every $P \in \mathcal{H}$,

$$\left|\mathbb{E}_P(F - c_G)^2 - V_G\right| \le C(\sqrt{M} + 1). \tag{27}$$

Taking the finite maximum gives $M \le V_G + C\sqrt{M} + C$. For $y = \sqrt{M}$, the positive quadratic root gives

$$y \le \frac{C}{2} + \sqrt{V_G + C + \frac{C^2}{4}} \le \sqrt{V_G} + C_1. \tag{28}$$

The maximum was finite independently of this inequality, so no desired hybrid variance bound has been assumed. Substituting (28) into (27) bounds the square comparison by $C_2(1 + \sqrt{V_G})$. Finally,

$$\operatorname{Var}_P F = \mathbb{E}_P(F - c_G)^2 - (\mathbb{E}_P F - c_G)^2.$$

The uniform hybrid mean comparison already proved bounds the subtracted term by a constant. This proves (4) and completes Theorem 1.1. □

5 Exact standardization and scope boundaries

Proof of Corollary 1.3. At fixed $\beta > 1$, Proposition 1.2 gives $V_G \sim a_2 n^{1/3} \to \infty$. By (4), uniformly over the permitted arrays,

$$\frac{\operatorname{Var} F(J)}{V_G} = 1 + O(V_G^{-1/2} + V_G^{-1}) \longrightarrow 1.$$

This proves the prefactor assertion and eventual positivity. In (5) take $a = \sqrt{V_G}$, which is at least one eventually, and $c = c_G$. For each fixed $t \in \mathbb{R}$, apply it to $\cos(tx)$ and $\sin(tx)$. All derivatives through order six are bounded. The characteristic functions of $(F(J) - c_G)/\sqrt{V_G}$ and $(F(g) - c_G)/\sqrt{V_G}$ differ by $O_t(V_G^{-1/2})$. Lévy's continuity theorem and (6) therefore give convergence of the former to ν_β. Equation (3) implies $(\mathbb{E}F(J) - c_G)/\sqrt{V_G} \to 0$. Slutsky's theorem with the variance ratio gives (7). Each estimate holds at every size and uniformly in the coordinate laws; no joint probability space across sizes or subsequence is needed. □

For $n = 2$ and symmetric signs $J_{12} \in \{-1, 1\}$, (1) is $2\log 2 + \log\cosh(q)$, a constant. Thus eventual positivity is the correct statement; no finite-size standardized variable is asserted when its variance vanishes. A nonzero external field would break (11), and is excluded. At $a = 1$ the smooth-test estimate is merely bounded, not a claim of convergence.

Under only a fourth moment, the pressure need not have all higher moments or exponential moments. The properties of the already defined limiting measure ν_β remain those supplied by the Gaussian input, but the proof does not transfer convergence of every finite-size moment. The assumptions permit asymmetric heavy tails with no fifth moment: for example, center and normalize a Pareto variable with tail $t^{-9/2}$ for $t \ge 1$. The proof uses none of the absent moments.

6 Provenance and formal support

The finite-size estimates form one integrated result. The variance-square algebra, restoration and finite-hybrid bootstrap received author-side assistance during pre-freeze feasibility; that assistance is recorded in the accompanying research provenance and is not represented as an independent proof review. The scope was fixed after feasibility work and retained through the subsequent proof and contribution reviews. Those reviews are internal project gates, not external peer review. A separate independent manuscript review is still required.

The primary comparison is bounded to the operations cited in Section 1. In particular, the EYY passage was read at the end of Section 6.1 and throughout Lemmas 6.5–6.6, not treated as an exhaustive survey of Green-function comparison. The complete precise reading record, input identities and source map accompany this manuscript. Third-party primary bodies are private review inputs and are not bundled in the manuscript payload.

Scoped formal support. The separate Lean package `o217-formal-v1` proves 28 theorems across 43 public support declarations. It constructs the actual finite zero-field spin model and vertex permutation, proves partition-function and log-partition gauge invariance, proves smoothness of the one-coordinate log partition, and derives oddness of the deleted cubic derivative, including observables of the log partition. It also proves the exact square identity and its fourth-order scalar remainder bound from supplied lower-order remainder inequalities; the scalar quadratic bootstrap and mean-square correction; and the edge count and both bounds in (15). The arbitrary-observable sign identity uses Lean's total derivative convention; it does not assert C^6 regularity or quantitative derivative bounds for an arbitrary observable.

The probability spaces and independent product laws, moment cancellation, Gibbs cumulant and mixed-derivative bounds, Taylor's theorem supplying the scalar remainder hypotheses, inner and outer replacements, integrability and deleted-coordinate restoration, and construction of the finite hybrid maximum remain human arguments. In particular, the bootstrap inequality is a premise of the scalar formal theorem; its probabilistic derivation is not formalized. The Gaussian source, characteristic-function argument, Lévy/Slutsky transfer and eventual positive variance also remain outside the formal coverage. Thus neither Theorem 1.1 nor Corollary 1.3 is claimed fully formalized. Exact declaration types and axiom reports accompany the sealed support package. The separate independent formal audit and its dependency bindings are recorded in the accompanying provenance.

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[5] OpenAI mathematics collection, *The low-temperature Sherrington–Kirkpatrick free-energy limiting law* (September 24, 2026), pinned revision `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, introduction, labels `eq:model`, `thm:main`, `thm:moments`. Stipulated source input. Exact pinned introduction.

[6] Same pinned manuscript as [5], Section 4, label `red:prop-moments`. Exact pinned reduction and completion removal.

[7] OpenAI mathematics collection, *The low-temperature Sherrington–Kirkpatrick fluctuation scale* (September 24, 2026), same pinned revision, introduction, label `thm:main`. Exact pinned scale theorem.